PROJECT REPORT

on

Email Fraud Detection Using

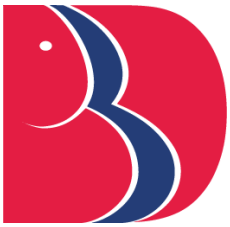
Machine Learning

Towards partial fulfillment of the requirements
for the award of degree of

Master of Computer Applications

from

Babu Banarasi Das University
Lucknow



Developed and Submitted by
Dipti Ranjan Sahoo

Under Guidance of
Mr. Nikhil Pandey

UNDER TAKING

This is to certify that Project Report entitled

Email Fraud Detection Using

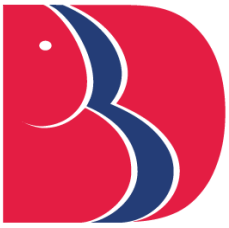
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Lucknow



Academic Year 2023-24

is a record of the student's own work carried out at

and to the best of our knowledge the work reported herein does not form a part of any other thesis or work on the basis of which degree or award was conferred on an earlier occasion to this or any other candidate.

Authorized Signatory
Organization Name

Students Signature

DECLARATION

I, hereby declare that the report entitled **" Email Fraud Detection Using Machine Learning"**, submitted to the **School of Computer Applications**, **Babu Banarasi Das University***, BBD City, Ayodhya Road, Lucknow, Uttar Pradesh – 226028*, is submitted in partial fulfilment of the requirements for the award of the degree of **Master of Computer Applications** in specialization with *Data Science & Artificial Intelligence.*

This report is the outcome of my own effort and has been completed under the guidance and supervision of **Mr. Nikhil Pandey** . I also declare that the contents of this report have not been submitted, either in full or in part, to any other university or institution for the award of any degree or diploma.

Place: Lucknow
Date:

Signature of the Student
Students' Name: Dipti Ranjan Sahoo
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Acknowledgement

I would like to express my sincere gratitude to all those who supported and guided me throughout the development of this project titled **"Email Fraud Detection Using Machine Learning."** This project has been a valuable learning experience, and its successful completion would not have been possible without the encouragement and assistance I received from various individuals.

First and foremost, I extend my heartfelt thanks to my project guide, **Mr. Nikhil Pandey**, for their constant support, expert guidance, and constructive feedback at every stage of this project. Their deep knowledge and experience in the domain of machine learning have been instrumental in shaping the direction and success of this work.

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I would like to acknowledge the support of my friends and peers who contributed through discussions, reviews, and testing, which helped improve the overall quality of the system. Their insights and encouragement kept me motivated throughout the project lifecycle.

Lastly, I express my deepest appreciation to my family for their unwavering support and encouragement, which gave me the strength and determination to complete this project successfully.

This project has helped me enhance my technical and problem-solving skills, and I am thankful to all who contributed to its successful completion.

INDEX

S. No	Title	Pg. No.
1.	Introduction	1-4
• 1.1	Background	1
• 1.2	Objective	1-3
• 1.3	Purpose & Scope	3-4
2.	Survey of Technology	5-8
3.	Requirement Analysis	9-21
• 3.1	Problem Definition	10
• 3.2	Planning & Scheduling	11
• 3.3	Requirement Specification	11-12
• 3.4	Preliminary Product Description	12-13
• 3.5	Conceptual Models System Architecture Design	13-21
4.	Coding	22-32
5.	Design Screenshots	33-36
6.	Testing	37-39
7.	Conclusion	40
8.	Future Scope	42
9.	Bibliography	43

1. INTRODUCTION

1.1 Background:

Nowadays,. Email fraud, including phishing, spoofing, and spam, is a major cybersecurity threat that targets individuals and organizations. Traditional rule-based systems often fail to detect sophisticated and evolving fraud techniques. As a result, machine learning (ML) has emerged as a powerful tool for automating and improving fraud detection in emails.

ML models can learn patterns from historical email data to distinguish between legitimate and fraudulent messages. These models analyze various features such as email content, sender information, subject lines, and embedded links. Techniques like Natural Language Processing (NLP) help understand text-based cues, while algorithms like decision trees, logistic regression, and neural networks enable accurate predictions.

Unlike static systems, ML models adapt over time, making them effective against new and unknown threats. Model performance is measured using metrics like accuracy, precision, and recall to ensure both high detection rates and low false positives. As email-based attacks become more complex, machine learning continues to play a critical role in enhancing digital communication security. In conclusion, email fraud detection using machine learning is a vital area of research and application that addresses the growing challenge of securing digital communication. With the continuous advancements in AI and data science, ML-based solutions are poised to become more accurate, scalable, and adaptive in the fight against email fraud.

1.2 Objective:

The main objective of this project is to develop a machine learning-based system that can accurately detect and classify fraudulent emails, such as phishing or spam, from legitimate ones. This includes:

- Collecting and preprocessing email datasets for training and evaluation.
- Extracting relevant features from email content and metadata using Natural Language Processing (NLP) techniques.
- Building and comparing different machine learning models to identify the most effective algorithm for fraud detection.
- Evaluating model performance using appropriate metrics like accuracy, precision, recall, and F1-score.
- Enhancing email security by minimizing false positives and improving the system's ability to detect new or evolving fraud patterns.

The ultimate goal is to create a scalable and adaptive solution that can assist users and organizations in reducing the risks associated with email-based attacks.

***System Objective*:** The objective of the system is to automatically detect fraudulent emails using machine learning techniques. It aims to distinguish between legitimate and malicious emails, such as phishing or spam, with high accuracy. The system will analyze features like email content, sender details, and embedded links. It will use classification algorithms and Natural Language Processing (NLP) to identify fraud patterns. The goal is to enhance email security by reducing user exposure to harmful emails.

System Context: The system operates in an environment where users regularly receive a mix of genuine and potentially harmful emails. It is designed to integrate with existing email platforms or operate as a standalone detection tool. The system processes incoming emails, extracts relevant data, and classifies them as legitimate or fraudulent. It interacts with users, email servers, and possibly a training dataset repository. The context includes both individual users and organizations looking to prevent cyber threats

Functional Requirement: The email fraud detection system must be capable of processing incoming email data, including sender details, subject lines, and body content. It should extract relevant features using Natural Language Processing (NLP) techniques such as tokenization, stopword removal, and keyword analysis. The system must then apply a trained machine learning model to classify emails as either legitimate or fraudulent. It should return the classification result along with a confidence score to help users assess the threat level. The system must also support training and evaluation using labeled datasets to ensure accuracy and reliability. Performance metrics like precision, recall, and F1-score should be used for evaluation. Optionally, a user interface may be provided to display emails and highlight suspicious ones. The system should handle a batch of emails efficiently and classify them in near real-time. Additionally, it may allow model updates with new data to adapt to emerging fraud techniques.

The functional requirements identified are:

a) **Email Input Processing**
   The system must accept and process raw email data, including the subject line, sender address, body content, and attachments if needed.
b) **Preprocessing and Feature Extraction**
   The system must clean the email text, remove stopwords, and extract relevant features (e.g., keyword frequency, URLs, sender domain) using NLP techniques.
c) **Fraud Detection Model**
   The system must implement machine learning algorithms (e.g., Logistic Regression, Decision Tree, or SVM) to classify emails as legitimate or fraudulent.
d) **Classification Output**
   The system must output the classification result, labeling each email as either "Fraudulent" or "Legitimate" with associated confidence/probability.
e) **Model Training and Evaluation**
   The system must allow training on labeled datasets and evaluate performance using metrics such as accuracy, precision, recall, and F1-score.

Non-Functional Requirement: . It is designed to integrate with existing email platforms or operate as a standalone detection tool. The system processes incoming emails, extracts relevant data, and classifies them as legitimate or fraudulent. It interacts with users, email servers, and possibly a training dataset repository.

a) Performance:
   The system should detect and classify emails in real-time or near real-time to ensure timely alerts and responses.
b) Scalability:
   The system must be able to handle a growing number of emails and users without a significant drop in performance.
c) Accuracy:
   The machine learning model should maintain a high accuracy rate (e.g., above 90%) with minimal false positives and false negatives.
d) Usability:
   The system interface should be user-friendly and intuitive, enabling users to view email classification results and alerts easily.
e) Security:
   The system must ensure data privacy and prevent unauthorized access to email content and user information.
f) Maintainability:
   The system should be modular and well-documented so that updates, such as retraining the model or improving the algorithm, can be done efficiently.
g) Reliability:
   The system must consistently provide correct and stable performance under normal and expected conditions.

1.3 Purpose and Scope

Purpose: The purpose of this project is to develop a machine learning-based system capable of detecting fraudulent emails, such as phishing, spam, and spoofing. As cyber threats continue to rise, individuals and organizations are increasingly at risk of email-based attacks. This system aims to enhance email security by automatically identifying suspicious messages and reducing human error..

This Vehicle Rental System is developed to provide the following services:

a) To develop a machine learning-based system for detecting fraudulent emails (phishing, spam, spoofing).

b) To improve email security by automatically identifying suspicious or malicious emails.

c) To reduce financial loss, identity theft, and data breaches caused by email fraud.

d) To utilize Natural Language Processing (NLP) and machine learning algorithms for accurate fraud detection.

e) ***Scope:*** The system will be able to classify emails as legitimate or fraudulent based on the learned patterns. The scope is limited to the detection of textual and structural email fraud; it does not include broader cybersecurity measures like malware detection or intrusion prevention. The project is intended for use in academic, personal, or small organizational environments.The area covers include:

- Collection and preprocessing of email datasets for training the model.
- Extraction of features from email text and metadata using NLP techniques.
- Implementation and comparison of machine learning algorithms like Logistic Regression, Random Forest, and SVM.
- Classification of emails into legitimate or fraudulent categories.
- Focus on textual and structural email fraud detection only (excludes malware or network security).
- Intended for academic research, personal use, and small organizational environments.

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2. SURVEY OF TECHNOLOGY

2.1 Introduction:

Email has evolved into a fundamental means of communication for both individuals and organizations. Its low cost, ease of use, and global reach have made it an indispensable tool in the digital world. However, the same attributes that make email useful also make it a common vector for cyberattacks. Fraudulent emails, including phishing, spoofing, spam, and business email compromise (BEC), pose significant threats to personal privacy, financial security, and corporate integrity. Traditional rule-based spam filters are often ineffective against sophisticated fraud techniques, necessitating the use of more advanced, data-driven approaches. Machine learning offers a dynamic solution by enabling systems to learn from patterns in data and make intelligent decisions. This survey explores the wide range of technologies used in building an email fraud detection system powered by machine learning.

Python Programming Language

Python is the cornerstone of this project, chosen for its simplicity, readability, and vast ecosystem of scientific libraries. As an interpreted, high-level programming language, Python allows rapid prototyping and iterative development. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming, making it adaptable for a variety of tasks. Python's integration with data analysis and machine learning libraries makes it particularly well-suited for fraud detection systems.

Key Python Libraries

- **NumPy**: Facilitates numerical operations and multi-dimensional arrays.
- **Pandas**: Provides data structures and functions needed for efficient data manipulation and analysis.
- **Matplotlib**: Used for data visualization to better understand distributions, correlations, and trends.
- **scikit-learn**: Offers robust tools for classification, regression, and clustering. It includes implementations of essential algorithms such as Logistic Regression, SVM, Random Forest, and Decision Trees.

Machine Learning Algorithms

Machine learning (ML) is the backbone of the fraud detection system. It enables the classification of emails into fraudulent and legitimate categories based on historical data. The following algorithms are frequently used in this domain:

- **Logistic Regression**: A simple yet effective binary classification algorithm ideal for linearly separable data.
- **Random Forest**: An ensemble learning method that improves prediction accuracy by aggregating multiple decision trees.
- **Support Vector Machine (SVM)**: Effective in high-dimensional spaces and suitable for cases where the number of dimensions exceeds the number of samples.
- **Naive Bayes**: Based on probability theory, it is particularly useful for text classification.

Evaluation metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are used to assess the performance of these models. Cross-validation techniques help prevent overfitting and ensure generalizability.

Natural Language Processing (NLP)

Since emails are composed of unstructured text, Natural Language Processing plays a vital role in extracting meaningful features. NLP allows machines to understand, interpret, and generate human language in a valuable way.

NLP Libraries and Techniques

- **NLTK (Natural Language Toolkit)**: Provides tools for tokenization, stemming, lemmatization, and stop word removal.
- **spaCy**: Offers industrial-strength NLP capabilities, including named entity recognition and dependency parsing.
- **TF-IDF Vectorizer**: Converts text data into numerical form by evaluating term frequency relative to its inverse document frequency.
- **Bag-of-Words (BoW)**: Represents text data as frequency counts of words.

These techniques allow the model to learn linguistic patterns associated with fraudulent behavior, such as urgency, threats, or requests for personal information.

Data Collection and Preprocessing

The foundation of any machine learning model is quality data. Public datasets like the Enron email dataset, SpamAssassin, or custom-scraped data sources are commonly used. The raw data undergoes several preprocessing steps:

- Removing HTML tags and scripts
- Normalizing text (lowercasing, removing punctuation)
- Tokenizing and lemmatizing words

- Extracting features such as subject lines, sender domains, and suspicious keywords

Feature engineering also includes the use of regular expressions to detect URLs, phone numbers, and email patterns commonly associated with fraud.

Django Web Framework

Django is a high-level Python web framework that promotes rapid development and clean, pragmatic design. It follows the Model-View-Template (MVT) architectural pattern and comes with a host of built-in features that streamline web application development.

Key Features of Django

- **ORM (Object Relational Mapping)**: Simplifies database interactions using Python classes.
- **Admin Interface**: Auto-generated interface for managing application data.
- **Form Handling**: Easy processing and validation of user input.
- **Middleware Support**: Allows for request and response processing layers.
- **Security**: Built-in protection against SQL injection, CSRF, XSS, and clickjacking.

Django is used in this project to create the frontend interface where users can input emails and receive classification results. It also manages backend processing, including calling the ML model for predictions.

Integration of Machine Learning Model

To integrate the trained ML model into Django, serialization tools such as pickle or joblib are used. The serialized model is loaded in Django views where prediction functions are triggered upon user input. The results are then displayed on the web interface in a user-friendly manner.

Django REST Framework (DRF)

For advanced use cases, Django REST Framework can be employed to convert the system into a RESTful API. This enables:

- Model access via HTTP requests
- Easy integration with mobile or third-party applications
- Support for authentication and permission controls

The use of DRF enhances the scalability and extensibility of the system.

Deployment Technologies

For deployment, the project can utilize platforms such as:

- **Heroku**: Simple cloud platform for hosting web apps

Software Development Tools

- **VS Code** : IDEs used for writing and debugging code.
- **Git & GitHub**: Version control and collaboration tools to manage project changes.

Evaluation:

For evaluation, the project uses standard machine learning metrics such as **accuracy**, **precision**, **recall**, **F1-score**, and **confusion matrix**. These metrics help determine how well the model identifies fraudulent emails without misclassifying legitimate ones. Precision and recall are particularly important in fraud detection, as a high false positive rate could cause inconvenience to users, while a high false negative rate could allow actual threats to go undetected. Cross-validation techniques are also applied to ensure the model's generalizability across different data samples.

Security Considerations

Given the sensitive nature of email data, security is paramount. Django offers built-in mechanisms to guard against common threats. In addition, input validation, HTTPS enforcement, encrypted storage, and secure authentication methods are implemented.

Conclusion

The survey of technology used in this project demonstrates the synergy between various tools and frameworks in building a comprehensive email fraud detection system. Python provides the programming backbone, machine learning models deliver intelligent predictions, NLP extracts meaningful features from unstructured text, and Django offers a robust web interface for user interaction. Together, these technologies form a scalable, secure, and effective solution to combat the ever-growing threat of email fraud.

3. REQUIREMENT ANALYSIS

3.1 Problem Definition

Email communication, despite being one of the most essential and efficient mediums for personal and business exchanges, remains a significant target for malicious actors. The evolution of digital technologies has not only improved communication but also increased the scope of fraudulent activities through emails. Attackers use techniques like phishing, spoofing, and scam emails to deceive users, steal sensitive information, and cause financial loss. Traditional filters based on fixed rules or blacklists have limited capabilities in adapting to the continuously evolving strategies of attackers. These systems often struggle with false positives and negatives, resulting in either missing malicious emails or marking legitimate messages as threats. The core problem lies in identifying these evolving threats dynamically and intelligently, which is where machine learning comes into play. Machine learning allows for pattern recognition and predictive modeling, offering an advanced method for fraud detection. Thus, the main problem addressed by this project is the development of a machine learning-based system that can effectively and accurately detect fraudulent emails using historical data and adaptive learning techniques.

3.2 Planning and Scheduling

Effective planning and scheduling are essential for the successful execution of this project. The planning stage began with a comprehensive analysis of the project requirements and scope. Tasks were broken down into distinct modules: requirement analysis, data acquisition, preprocessing, model training, integration, interface development, testing, and deployment. Each task was allocated a specific time frame and responsible team members. Planning tools like Gantt charts were employed to visualize timelines, deadlines, and dependencies. Agile methodology was adopted, emphasizing iterative progress and continuous feedback. Weekly sprints and review meetings ensured that all deliverables remained on schedule. Resources were managed efficiently, and tools like GitHub facilitated version control, while platforms like Trello helped in task tracking and team coordination. Time buffers were included to accommodate unforeseen delays or technical challenges. This structured planning approach ensured smooth workflow, reduced bottlenecks, and promoted timely delivery of all components of the system.

3.3 Requirement Specification

Requirement specification outlines both the functional and non-functional expectations of the system.

Functional Requirements:

- Allow users to input email content through a web interface.
- Classify emails as "Fraudulent" or "Legitimate."
- Maintain logs of all classifications.
- Enable administrators to retrain models using updated datasets.
- Provide instant prediction results with detailed output.

Non-Functional Requirements:

- High accuracy (over 90%) of predictions.
- Response time under 2 seconds.
- Secure handling of user data and predictions.
- Scalable system architecture.
- Compatibility across browsers and mobile platforms.

System Requirements:

- Programming Language: Python 3.10+
- Framework: Django for web development
- Libraries: NumPy, Pandas, scikit-learn, joblib, NLTK, spaCy
- Database: SQLite or PostgreSQL
- Deployment: Heroku or AWS

This specification ensures clarity between developers and stakeholders, acting as a blueprint for system design and development.

Hardware Requirements

• Pentium IV Processor

• 512 MB RAM

• 40 GB HDD

3.4 Preliminary Product Description

The preliminary product is a responsive web-based platform that allows users to input email text and get an instant classification result. It includes a user-friendly frontend built using Django templates, where users can paste email content. The backend processes this text using NLP techniques, extracts features using TF-IDF, and classifies the email using a pre-trained machine learning model. The result is shown immediately, with a clear indication of the email being either safe or suspicious. An admin dashboard provides tools for monitoring performance, uploading new datasets, retraining the model, and viewing analytics. The product emphasizes ease of use, fast response times, and high accuracy. It is built with modular components, allowing for future integration with enterprise systems or mobile apps.

3.5 Conceptual Models System Architecture Design

Defining A System: Defining a system is a critical initial phase that sets the stage for the entire development process. This phase involves conceptualizing, designing, and describing the software solution that will address the identified problem or need. It serves as a roadmap for the development team and stakeholders, guiding the subsequent stages of the SDLC. Defining a system encompasses several key aspects, each crucial for the success of the project. First and foremost, defining a system involves understanding and articulating the problem or need that the software aims to address. This requires collaboration among stakeholders, including clients, end users, project sponsors, and domain experts. Through discussions, interviews, surveys, and other techniques, stakeholders identify the pain points, challenges, and opportunities that the software solution should target. This process of problem definition provides the context and motivation for the development effort, ensuring that the resulting system meets the real-world needs of its users. Once the problem is defined, the next step is to establish the scope of the system. The scope defines the boundaries of what the software will and will not do, outlining the 13 functionalities, features, and components that will be included in the final product. Stakeholders prioritize requirements based on their importance, urgency, and feasibility, helping to define a clear scope for the project. This ensures that the development team stays focused on delivering value to the users while managing expectations and avoiding scope creep. With the scope defined, attention turns to the architecture of the system. System architecture outlines the high-level structure and design of the software solution, including the arrangement of components, modules, and subsystems, as well as the interactions and interfaces between them. It provides a blueprint for developers to follow during the implementation phase, guiding decisions about technology stack, frameworks, and design patterns. A well-designed architecture lays the foundation for a scalable, maintainable, and extensible system, capable of evolving with changing requirements and technologies. In parallel with defining the architecture, the system's data model is designed. The data model describes the organization and structure of the data that the system will manage,

including data entities, attributes, relationships, and constraints. Designing a data model ensures that the system can effectively store, retrieve, and manipulate data to meet the requirements of the users. It lays the groundwork for database design, data storage, and data access mechanisms, ensuring data integrity, consistency, and security. Another crucial aspect of defining a system is designing the user interface (UI). The UI design focuses on creating intuitive and visually appealing interfaces that enable users to interact with the system. This involves designing screens, forms, menus, and other elements to facilitate user input and feedback. Lastly, defining a system involves specifying both functional and non-functional requirements. Functional requirements describe the specific actions and behaviors that the system must perform to fulfill its intended purpose. These requirements define the functionalities and features that users expect from the software, such as data processing, calculations, reporting, and integration with other systems. Non-functional requirements address the quality attributes of the system, such as performance, reliability, security, scalability, and maintainability. These requirements define the criteria for evaluating the system's overall effectiveness and usability beyond its functional capabilities. Input Output The architecture is based on a three-tier structure:

- **Presentation Layer:** The frontend built using Django templates, HTML, CSS, and JavaScript. This layer interacts with users and handles data input and output.
- **Logic Layer:** This is the backend engine powered by Django views and Python scripts. It includes NLP preprocessing, feature extraction, and classification modules.
- **Data Layer:** Responsible for storing email records, prediction logs, model files, and user session data. SQLite is used in development, with potential for PostgreSQL in production.

The modular architecture ensures separation of concerns and allows for easy debugging, scaling, and upgrading.

3.6 System Development Cycle

The System development life cycle (SDLC), or Software development processing systems engineering, information systems and software engineering, is a process of creating or altering information systems, and the models and methodologies that people use to develop these systems. In software engineering, the SDLC concept underpins many kinds of software development methodologies. It provides a structured framework for managing the entire software development process, from the initial conception of an idea to the final delivery of a functional product. SDLC consists of several phases, each with its specific objectives, activities, and deliverables, ensuring that the software meets quality standards, user requirements, and business goals. These methodologies form the framework for planning and controlling the creation of an information system the process. Broadly, following are the different activities to be considered while defining

the system development life cycle for the said project:

• Problem Definition

• System Analysis

• Study of existing system

• Drawback of the existing system

• Proposed system

• System Requirement study

• Data flow analysis

• Feasibility study

• System design

• Input Design (Database & Forms)

• Updating

• Query /Report design

• Administration

• Testing

• Implementation

• Maintenance

3.7 System Analysis

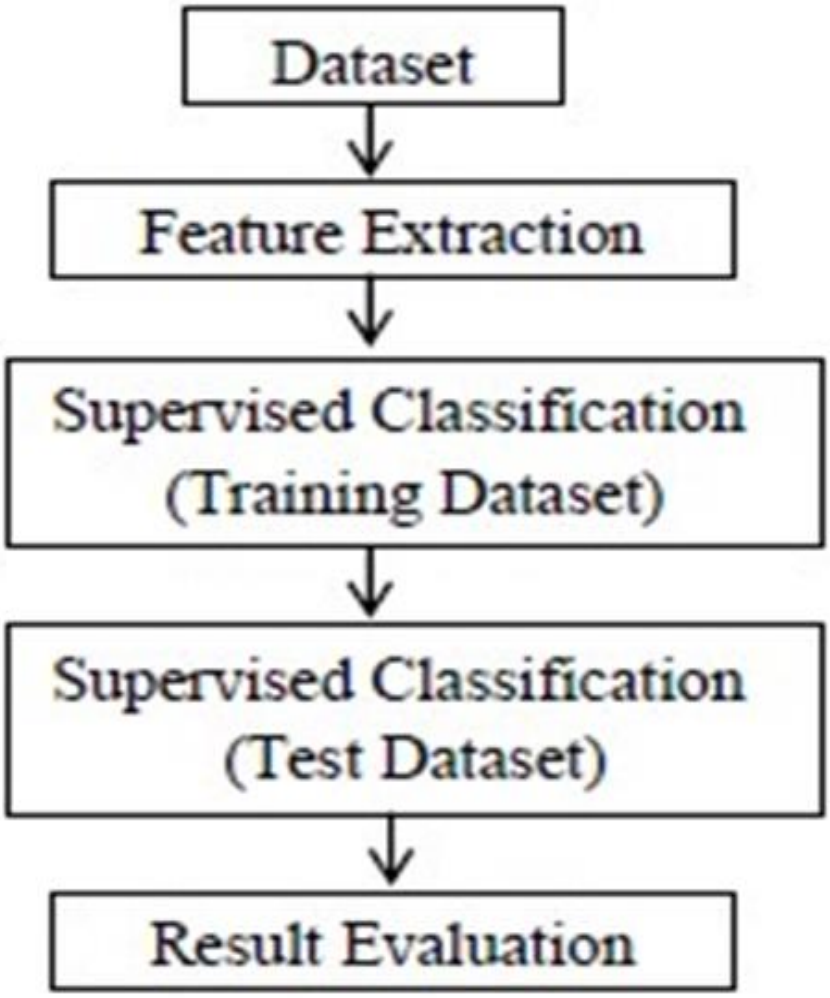
System analysis is a critical phase in the software development life cycle (SDLC) that involves studying, understanding, and defining the requirements of a proposed software system or application. It is a systematic approach to investigating, identifying, and documenting the needs, objectives, and constraints of the system to be developed. System analysis serves as the foundation for the subsequent phases of software development, providing the necessary insights and information for designing, implementing, and testing the system effectively. The process of system analysis begins with gathering information from stakeholders, 15 including end-users, clients, managers, and domain experts, to gain a comprehensive understanding of the problem domain and the goals of the proposed system. This often involves conducting interviews, surveys, workshops, and other techniques to elicit requirements and capture stakeholders' perspectives on

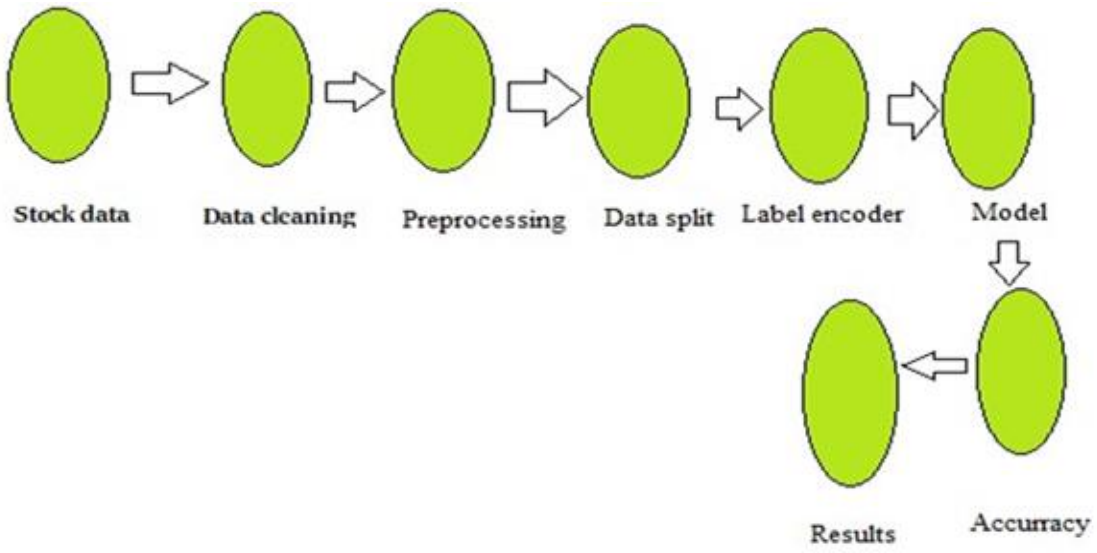
what the system should accomplish. Requirements gathering is a collaborative effort that aims to identify both functional and non-functional requirements, as well as any constraints or dependencies that may impact the design and implementation of the system. Once the requirements have been collected, system analysts analyze and prioritize them to determine their significance and feasibility. This involves evaluating the scope of the project, assessing the potential risks and benefits, and identifying any conflicting or ambiguous requirements that may need further clarification. Requirements analysis requires careful attention to detail and a deep understanding of the domain in which the system will operate, as well as consideration of factors such as cost, schedule, and available resources. During the analysis phase, system analysts also work to develop a conceptual model of the system that captures its key components, processes, and interactions. This may include creating diagrams, such as data flow diagrams, use case diagrams, or entity-relationship diagrams, to represent the structure and behavior of the system from a high-level perspective. The conceptual model serves as a visual aid for stakeholders to understand the proposed system and provides a basis for further refinement and elaboration during the design phase. Throughout the analysis phase, system analysts collaborate closely with stakeholders to validate requirements, gather feedback, and ensure that the proposed solution aligns with the needs and expectations of its users. This may involve conducting reviews, demonstrations, or prototypes to solicit input and verify that the system will meet its intended objectives. Effective communication and collaboration are essential for building consensus and ensuring that all stakeholders have a shared understanding of the system requirements and design. In summary, system analysis is a systematic and collaborative process of studying, understanding, and defining the requirements of a software system or application. It involves gathering information from stakeholders, analyzing and prioritizing requirements, developing a conceptual model of the system, evaluating alternative solutions, and making recommendations for the best course of action. By providing a clear understanding of the problem domain and the goals of the proposed system, system analysis lays the groundwork for the successful development and implementation of software solutions that meet the needs of their users.

3.8 System Design

System design is a comprehensive process that encompasses the creation of a blueprint for the development and implementation of a software system or application. It involves translating requirements gathered during the initial stages of software development into 16 a detailed plan that outlines how the system will function, how its components will interact, and how it will be built and deployed. System design is a crucial phase in the software development life cycle (SDLC) as it sets the foundation for the construction and operation of the final product. At its core, system design involves making a series of decisions about the architecture, components, modules, interfaces, and data structures of the system. These decisions are guided by the functional and non-

functional requirements of the system, as well as considerations such as scalability, reliability, performance, security, and maintainability. System designers must strike a balance between meeting these requirements while also ensuring that the system is feasible to implement within the constraints of time, budget, and available technology. The first step in system design is to analyze and understand the requirements of the system. This involves gathering information from stakeholders, including end-users, clients, and domain experts, to identify the goals, objectives, and constraints of the system. Requirements analysis helps to define the scope of the system and establish a clear understanding of what needs to be accomplished. Once the requirements are understood, system designers begin the process of conceptualizing the system architecture. This involves defining the overall structure of the system, including its high-level components, modules, and their interactions. Architecture design focuses on identifying the key subsystems and their responsibilities, as well as defining the interfaces between them. At this stage, designers may use techniques such as architectural patterns, such as client-server, layered, or microservices architecture, to organize and structure the system effectively. With the architecture in place, system designers move on to detailed design, where they flesh out the internal workings of each component and module. This involves specifying the algorithms, data structures, and logic necessary to implement the system's functionality. Design decisions at this level are guided by principles of modularity, encapsulation, and information hiding, which aim to promote reusability, maintainability, and flexibility. Once the design is complete, system designers document their decisions and create detailed design specifications that serve as a guide for developers during the implementation phase. These specifications may include architectural diagrams, data models, interface definitions, algorithms, and other design artifacts. Clear and comprehensive documentation is essential for ensuring that the system is implemented correctly and that all stakeholders have a shared understanding of its design and functionality. Finally, system designers may also conduct design reviews and evaluations to validate the design against the requirements and identify any potential issues or improvements. Iterative refinement of the design may be necessary as new information becomes available or as the project 17 progresses. Ultimately, the goal of system design is to create a robust, scalable, and maintainable solution that meets the needs of its users and stakeholders. System design translated functional requirements into a technical blueprint. The frontend was designed for simplicity and usability, while the backend focused on efficiency and modularity. NLP components were integrated for text preprocessing, and the machine learning model was encapsulated within a prediction module.


Dataset
Feature Extraction
Supervised Classification
(Training Dataset)
Supervised Classification
(Test Dataset)
Result Evaluation


Stock data
Data cleaning
Preprocessing
Data split
Label encoder
Model
Results
Accurracy

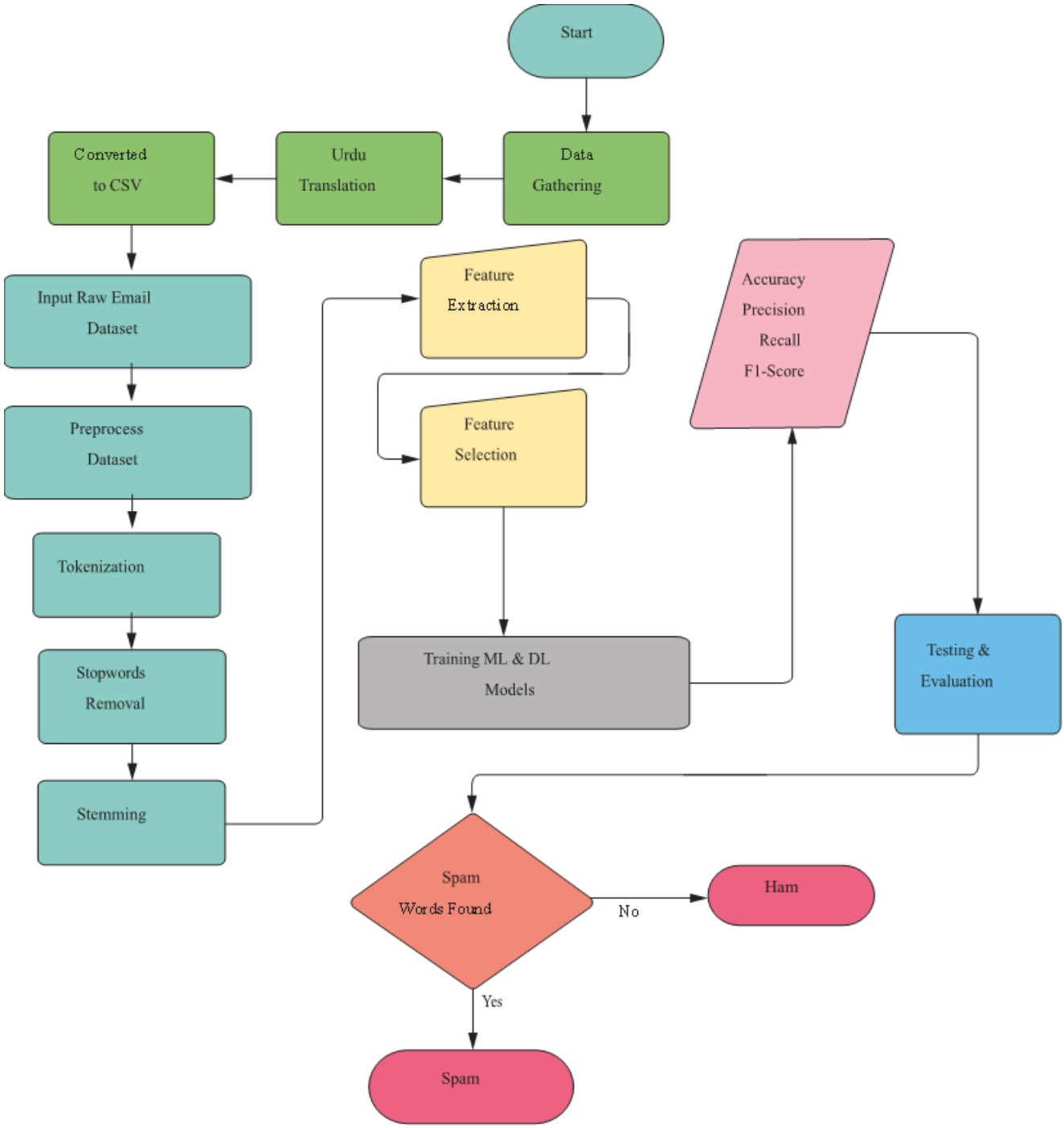
3.9 Entity-Relationship Diagram (ERD)

The Entity Relation Model or Entity Relation Diagram (ERD) is a data model or diagram for high-level description of conceptual data model, and it provides a graphical notation for representing such data models in the form of entity relationship diagrams. Such models are typically used in the first stage of Management information system design; they are used for example, to describe information needs and/ or the type of information that is to bestored in the Database during the requirement analysis. The data modeling technique, however, can be used to describe any ontology (i.e. an overview and classification of used term and their relationships) for a certain universe of discourse (i.e. area of interest). At the heart of an ER diagram are entities, which are objects or concepts with independent existence and propertiesthat are relevant to the database. Entities can represent real-world objects like people, places, or things, or they can be abstract concepts like accounts or transactions in a banking system. Each entity is depicted as a rectangle in the ER diagram, labeled with its name.

Attributes are the properties or characteristics of entities that are stored in the database. These attributes describe the features of an entity and are represented as ovals connected to their respective entities by lines. For example, in a database for a university, a "Student" entity may have attributes such as "Student ID," "Name," "Date of Birth," and "Major." Relationships define how entities interact with each other within the database. They represent the associations between entities and are crucial for understanding the connections between different parts of thesystem. Relationships are illustrated as lines connecting entities, typically labeled with verbs or phrases that describe the nature of the association. For instance, in a library database, a "Borrower" entity may have a relationship with a "Book" entity labeled as "Borrows," indicating that a borrower can borrow multiple books. To represent a many-to-many relationship in an ER diagram, it is common to introduce a junction entity, also known as an associative entity or a linking table. This junction entity resolves the many-to-many relationship into two one-to-many relationships, simplifying the structure of the database. Keys are crucial for maintaining data integrity and ensuring efficient database operations, while cardinality constraints help define the nature of the relationships between entities.Overall, ER diagrams provide a powerful visual tool for database designers to analyze, design, and communicate the structure of a database system.By representing entities, attributes, and relationships in a clear and concise manner, ER diagrams facilitate collaboration between stakeholders and ensure that the database meets the requirements of its users.

The ERD maps the relationships between system entities:

- **User:** user_id, name, email, password
- **Email_Record:** record_id, user_id, content, timestamp
- **Prediction:** prediction_id, record_id, result, confidence



- **Admin:** admin_id, username, password

Relationships:

- One User can have multiple Email_Records.
- One Email_Record has one Prediction.
- Admin oversees Email_Records and Model Training.This structure ensures relational integrity and supports analytics.

3.10 Data Flow Diagram (DFD)

The data flow diagram shows the flow of data within any system. It is an important tool for designing phase of software engineering. Larry Constantine first developed it. It represents graphical view of flow of data. It's also known as BUBBLE CHART. The purpose of DFD is major transformation that will become in system design symbols used in DFD.

In the DFD, four symbols are used and they are as follows.

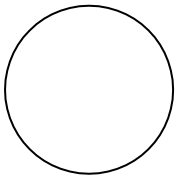
- A square defines a source (originator) or destination of system data.



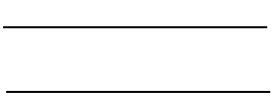
- An arrow identifies data flow-data in motion. It is 2a pipeline through which information flows.



- A circle or a "bubble "(Some people use an oval bubble) represents a process that transfers informing data flows into outgoing data flows.



- An open rectangle is a data store-data at rest, or a temporary repository of data.



The general DFD represents the overall data movement:

- User inputs email content
- System preprocesses the email
- Feature vectors are generated
- Prediction is made and stored
- Output is returned to the user

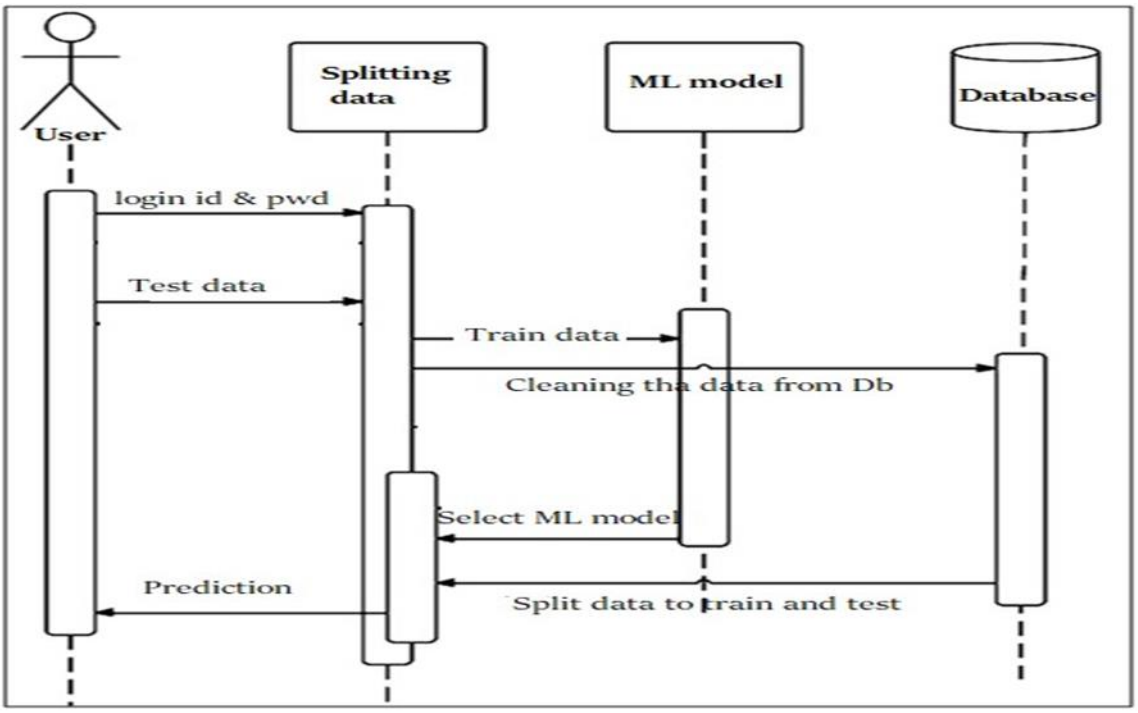
It helps visualize system operation and guides development.

3.11 Context Level Data Flow Diagram

: This level shows the overall context of the system and its operating environment and shows the whole system as just one process. Online book store is shown as one process in the context diagram; which is also known as zero level DFD, shown below. The context-level DFD depicts the system as a single process:

- **External Entities:** User, Admin
- **Processes:** Email input, Model Prediction
- **Data Stores:** Email Records, Prediction Logs
- **Data Flows:** User data input, prediction results, model updates

This diagram shows high-level interactions and data movement between actors and the system.



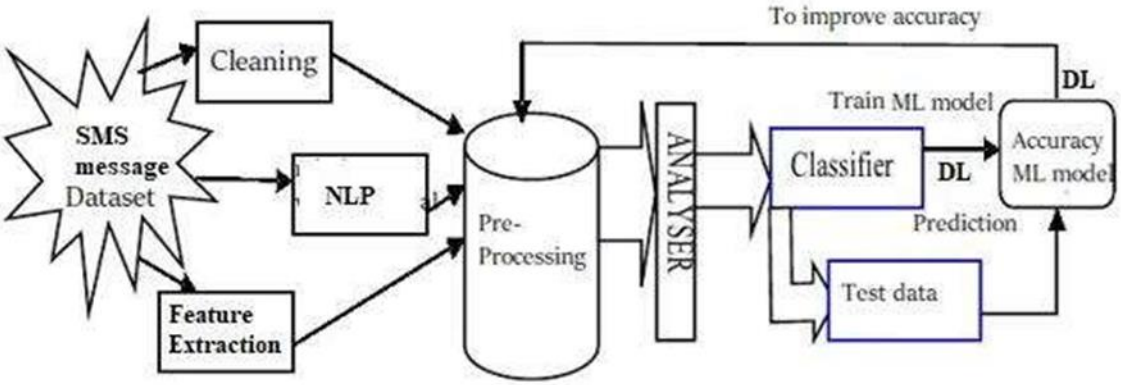
3.12 Zero Level Data Flow Diagram

The context diagram plays important role in understanding the system and determining the boundaries. The main process can be broken into sub-processes and system can be studied with more detail; this is where 1st level DFD comes into play. This DFD breaks down the main process into sub-processes:

- **P1:** Accept Email Input
- **P2:** Preprocess Email Content
- **P3:** Generate Features
- **P4:** Predict Using ML Model

- **P5:** Store Results in DB
- **P6:** Display Result to User

Data stores and flows are connected to these processes, showing how input transforms to output.



4. CODING

4.1 : myModel.pkl

Coding:

```
import numpy as np
import pandas as pd
import joblib
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer

df = pd.read_csv('spam.tsv', sep="\t")

hamDf = df[df['label'] == "ham"]
spamDf = df[df['label'] == "spam"]

hamDf = hamDf.sample(spamDf.shape[0])

finalDf = pd.concat([hamDf, spamDf], ignore_index=True)



X_train, X_test, Y_train, Y_test = train_test_split(finalDf['message'], finalDf['label'], test_size = 0.2,
random_state = 0, shuffle = True, stratify=finalDf['label']
                                        )
# Pipeline 1
model = Pipeline([
    ('tfidf', TfidfVectorizer()),
    ('clf', RandomForestClassifier(n_estimators=100, n_jobs=-1))
])



model.fit(X_train,Y_train)

Y_pred = model.predict(X_test)
```

```
print(accuracy_score(Y_test, Y_pred))


joblib.dump(model,"myModel.pkl")
```

4.2 : mySVCModel.pkl

```
# Pipeline 2
model = Pipeline([
    ('tfidf', TfidfVectorizer()),
    ('model', SVC(C=1000, gamma = 'auto'))
])



model.fit(X_train,Y_train)


Y_pred = model.predict(X_test)


print(accuracy_score(Y_test, Y_pred))


joblib.dump(model,"mySVCModel.pkl")
```

4.3 : views.py

```
from django.shortcuts import render
import os
import joblib

# Load models
model1_path = os.path.join(os.path.dirname(__file__), 'myModel.pkl')
model2_path = os.path.join(os.path.dirname(__file__), 'mySVCModel.pkl')

model1 = joblib.load(model1_path)
model2 = joblib.load(model2_path)

def index(request):
    return render(request, 'index.html')

def checkSpam(request):
    if request.method == "POST":
        algo = request.POST.get("algo")
        rawData = request.POST.get("rawdata")
```

```
        if not rawData:
            return render(request, 'index.html', {"error": "Input cannot be empty"})

        try:
            if algo == "Algo-1":
                finalAns = model1.predict([rawData])[0]
            elif algo == "Algo-2":
                finalAns = model2.predict([rawData])[0]
            else:
                finalAns = "Invalid algorithm selection"
        except Exception as e:
            finalAns = f"Error during prediction: {str(e)}"

        return render(request, 'output.html', {"answer": finalAns})
    else:
        return render(request, 'index.html')
```

4.4 manage.py

```
#!/usr/bin/env python

Django's command-line utility for administrative tasks.
import os
import sys


def main():
 Run administrative tasks.
    os.environ.setdefault('DJANGO_SETTINGS_MODULE', 'EFD.settings')
    try:
        from django.core.management import execute_from_command_line
    except ImportError as exc:
        raise ImportError(
            "Couldn't import Django. Are you sure it's installed and "
            "available on your PYTHONPATH environment variable? Did you "
            "forget to activate a virtual environment?"
        ) from exc
    execute_from_command_line(sys.argv)


if __name__ == '__main__':
    main()
```

4.7 urls.py

```
from django.contrib import admin
from django.urls import path
from . import views

urlpatterns = [
    path('', views.index, name="Index"),
    path('check', views.checkSpam, name="CheckSpam"),  # <-- fix here
```

4.5 index.html

```
<!DOCTYPE html>
<html lang="en">
{% load static %}
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.6/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
4Q6Gf2aSP4eDXB8Miphtr37CMZZQ5oXLH2yaXMJ2w8e2ZtHTl7GptT4jmndRuHDT"
crossorigin="anonymous">
   <link rel="stylesheet" href="{% static 'Css/Style.css' %}">
   <link rel="stylesheet" href="{% static 'Spam/Css/bootstrap.min.css'%}">
   <link rel="stylesheet" href="{% static 'Spam/Css/Style.css' %}">
   <link rel="stylesheet" href="{% static 'Spam/Css/radio.css' %}">
   <title>AI Tools | NLP - Email Fraud Detection</title>
   <meta http-equiv="X-UA-Compatible" content="ie=edge">
</head>

<body>
   <section>
      <header>
         <nav>
            <div><a href="#" class="brand">EMAIL FRAUD DETECTION</a></div>
            <ul>
               <li><a href="#" class="active">Web App</a></li>

            </ul>
         </nav>
      </header>
      <div class="fluid-container content">
         <div class="row m-auto">
            <div class="col-md-6 p-5 m-auto typo-content">
               <h1>
                  welcome
                  .
                  .
               </h1>
               <p>
                  This is an online platform where
                  you can check any text to determine if
                  it is a Spam Message Text or not.
               </p>
               <p>
                  Just enter your text into the
                  Textarea and select an algorithm,
                  press the CHECK button and wait
                  while your data is processed.
               </p>
            </div>
            <div class="col-md-6 p-5 m-auto form-content">
               <form method="post" action="check" class="main-form">
                  {% csrf_token %}
                  <textarea name="rawdata" placeholder="Enter Your Text Here..."
```

```
id="rawdata"></textarea><br>
                    <div class="algo-choice">
                        <label class="radiocontainer">Algo - 1
                            <input type="radio" name="algo" value="Algo-1">
                            <span class="checkmark"></span>
                        </label>
                        <label class="radiocontainer">Algo - 2
                            <input type="radio" name="algo" value="Algo-2">
                            <span class="checkmark"></span>
                        </label>

                    </div>
                    <div class="check-holder">
                        <button type="submit" class="check-btn">Check</button>
                    </div>
                </form>
            </div>
            </div>
        <div class="footer">
            </div>
        </div>
    </section>
    <script src="https://cdnjs.cloudflare.com/ajax/libs/jquery/3.7.1/jquery.js"></script>
    <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.6/dist/js/bootstrap.bundle.min.js"
integrity="sha384-
j1CDi7MgGQ12Z7Qab0qlWQ/Qqz24Gc6BM0thvEMVjHnfYGF0rmFCozFSxQBxwHKO"
crossorigin="anonymous"></script>
    <script src="{% static 'Js/Main.js' %}"></script>
</body>

</html>
```

4.6 Style.css

```
@import 'fonts.css';

* {
    padding: 0;
    margin: 0;
    box-sizing: border-box;
}

section {
    width: 100%;
    min-height: 100vh;
    background: url('../Images/bg.jpg');

    background-size: cover;
    background-repeat: no-repeat;
}

header {
    width: 100%;
    height: 3.5em;
```

```
    position: fixed;
    top: 0;
    left: 0;
    z-index: 10;
    background: rgba(0, 0, 0, 0.5);
}

header nav {
    height: 3.5em;
    margin-right: 2em;
    display: flex;
    justify-content: space-between;
}

header nav .brand {
    font-size: 2.3rem;
    color: #fff;
    font-family: Jonesy, sans-serif;
    text-decoration: none;
    padding: 0 0 0 2rem;
}

header nav ul {
    list-style: none;
    display: flex;
}

header nav ul li a {
    display: block;
    padding: 0.6em 1em;
    text-decoration: none !important;
    color: gray !important;
    font-size: 1.3rem;
    font-family: Azonix, sans-serif;
    transition: 0.3s color;
}

header nav ul li a:hover {
    color: #fff !important;
}

.active {
    color: #fff !important;
}

.content {
    padding: 0 2em;
    padding-top: 3.5em;
    min-height: 100vh;
    display: flex;
    justify-content: center;
    align-items: center;
    position: relative;
}
```

```
.main-form textarea {
    width: 100%;
    background: rgba(0, 0, 0, 0.5);
    outline: none;
    border: 2px white solid;
    border-radius: 0.3em;
    position: relative;
    height: 10em;
    margin-bottom: 0.5em;
    resize: vertical;
    padding: 0.5em;
    color: #fff;
    font-family: RobotL, sans-serif;
    font-size: 1.3rem;
}

.content h1,
.content h5,
.content p {
    color: #fff;
}

.content h1 {
    font-family: Azonix, sans-serif;
    margin: 0;
    padding: 0;
}

.content h5 {
    margin-top: 0;
    padding-top: 0;
    font-family: Azonix, sans-serif;
}

.content p {
    font-family: Centaur, serif;
    font-size: 1.7rem;
    line-height: 1.7rem;
}

.algo-choice {
    width: 100%;
    display: flex;
    flex: 1;
    justify-content: space-between;
}

.radiocontainer {
    display: block;
    position: relative;
    padding-left: 35px;
    margin-bottom: 12px;
    color: #fff;
    -webkit-user-select: none;
    -ms-user-select: none;
```

```
    user-select: none;
}

.radiocontainer input {
    position: absolute;
    top: 0;
    left: 0;
}

.checkmark {
    position: absolute;
    top: 0;
    left: 0;
    height: 25px;
    width: 25px;
    background: #fff;
    border-radius: 10%;
}

.radiocontainer:hover input ~ .checkmark {
    background: #ccc;
}

.radiocontainer input:checked ~ .checkmark {
    background-color: gray;
}

.checkmark:after {
    content: '';
    position: absolute;
    display: none;
    top: 7px;
    left: 7px;
    width: 10px;
    height: 10px;
    border-radius: 10%;
    background: #fff;
}

.radiocontainer input:checked ~ .checkmark:after {
    display: block;
}

.check-holder {
    display: flex;
    justify-content: end;
}

.check-btn {
    padding: 0.4em 4em;
    border: none;
    border-radius: 0.3em;
    background: #fff;
    font-family: Roboto, sans-serif;
    font-size: 1.5rem;
```

```
    cursor: pointer;
}

.footer {
    position: absolute;
    bottom: 0;
}

.footer p {
    font-size: 1.2rem;
    font-family: Centaur, serif;
    letter-spacing: 1px;
    text-align: center;
}

/* Answer Page Style */

.AnswerSection {
    width: 1000px;
    height: 100vh;
    display: flex;
    flex-direction: column;
    justify-content: center;
    align-items: center;
}

.AnswerSection .answer {
    color: white;
    font-size: 5rem;
    text-align: center;
}

.backbtn,
.backbtn:hover {
    padding: 0em 1em;
    color: #fff;
    font-size: 3rem;
    text-decoration: none;
    border: 5px #fff solid;
    border-radius: 0.2em;
}

@media only screen and (max-width: 768px) {
    .form-content {
        padding-top: 0;
    }

    .main-form {
        margin-bottom: 2em;
    }

    .typo-content {
        padding-bottom: 0 !important;
    }
}
```

```
@media only screen and (max-width: 410px) {
   header nav {
      margin-right: 1em;
   }

   header nav ul li a {
      padding: 1em 0.4em;
      font-size: 1rem;
   }
}
```

4.7. output.html

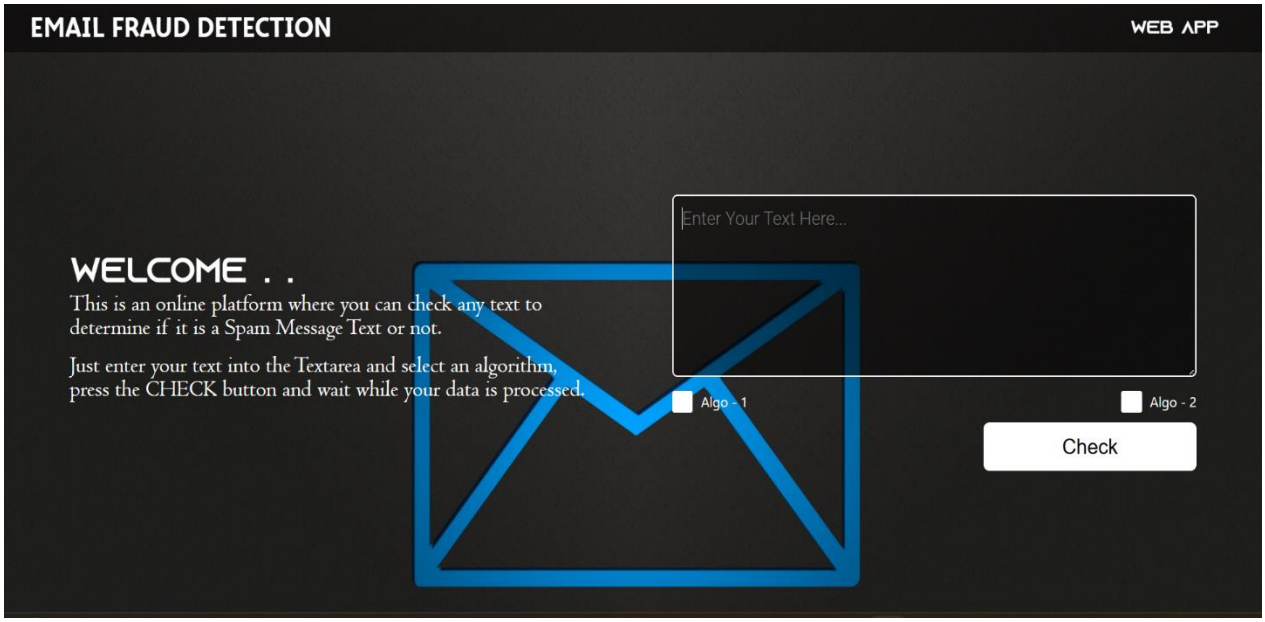
```
<!DOCTYPE html>
<html lang="en">
{% load static %}
<head>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
   <meta http-equiv="X-UA-Compatible" content="ie=edge">
   <link rel="stylesheet" href="{% static 'Spam/Css/bootstrap.min.css'%}">
   <link rel="stylesheet" href="{% static 'Spam/Css/Style.css' %}">
   <link rel="stylesheet" href="{% static 'Spam/Css/radio.css' %}">
   <title>Document</title>
</head>

<body>
   <!-- Main Section Start -->
   <section class = "AnswerSection">
      <p class = "answer">
         {% if answer == "ham" %}
         Not Spam
         {% else %}
         Spam
         {% endif %}
       </p>
      <a href="/" class = "backbtn">Back</a>
      </section>
   <!-- Main Section End -->
   <script src="./Js/JQuery.js"></script>
   <script src="./Js/Bootstrap.min.js"></script>
   <script src="./Js/Main.js"></script>
</body>

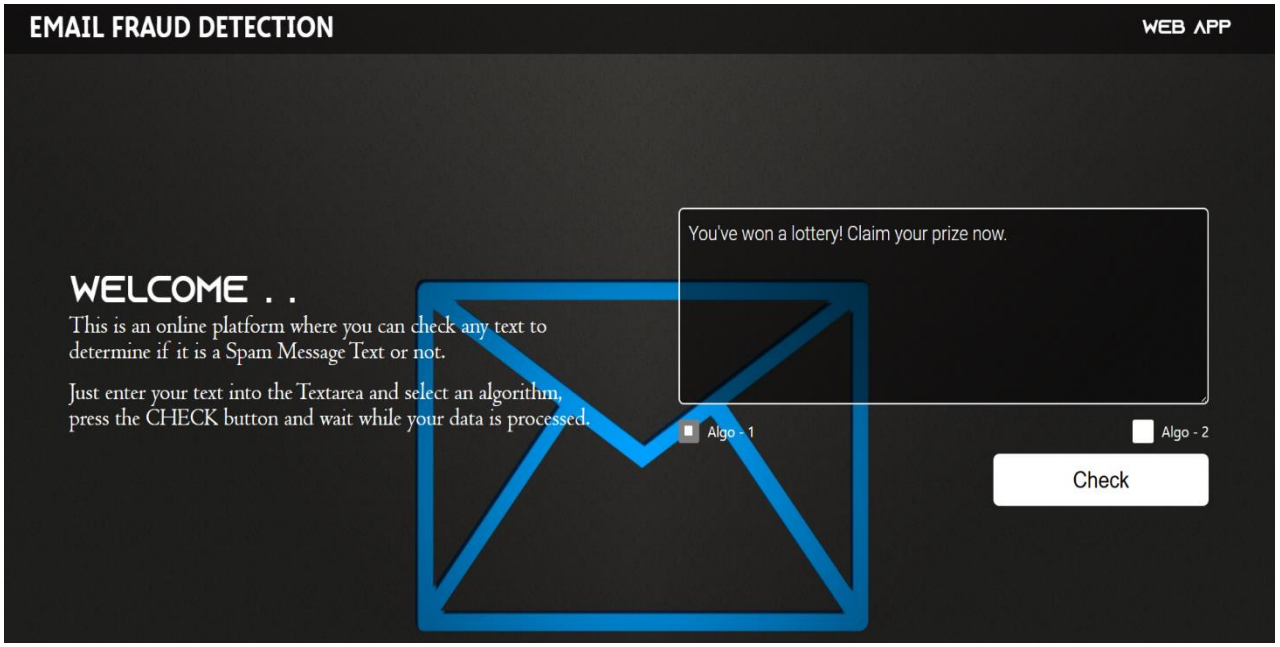
</html>
```

5. DESIGN SCREENSHOTS

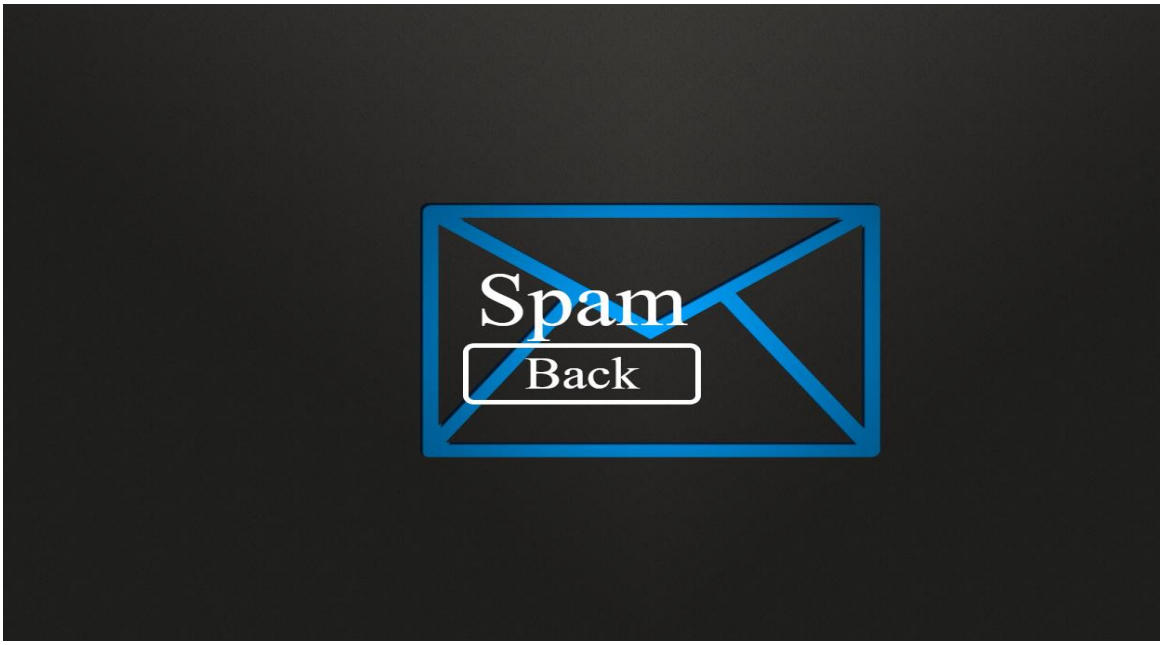
Login Window:



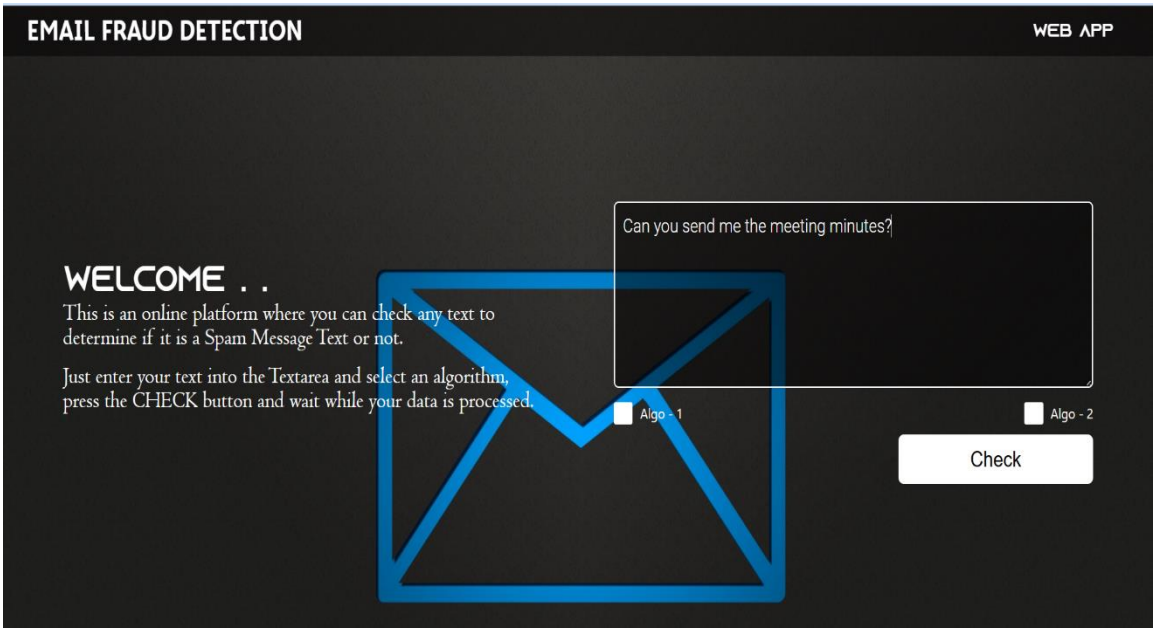
For checking spam message - You've won a lottery! Claim your prize now.



output



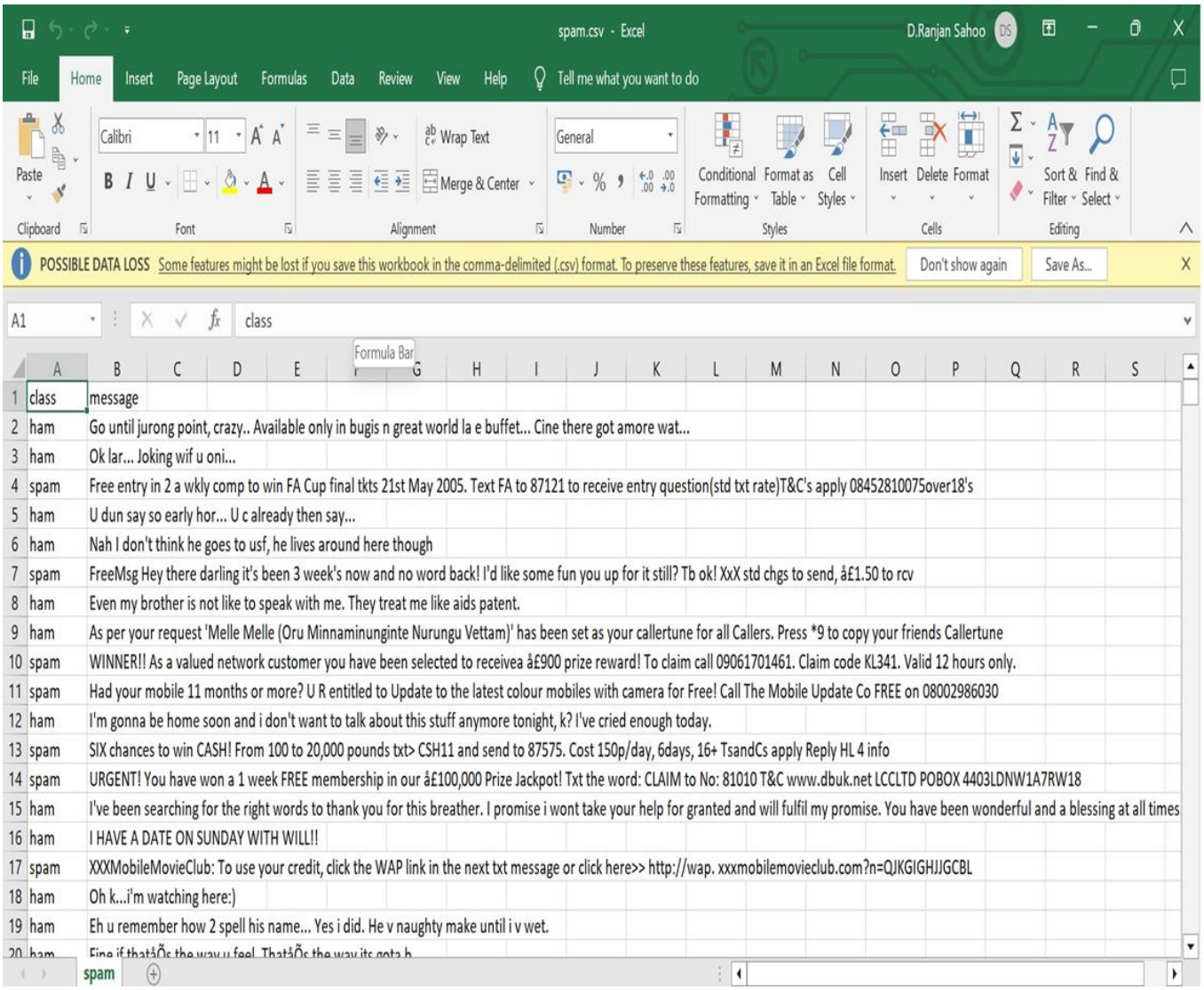
For checking ham message - Can you send me the meeting minutes?



output



Spam.csv file



spam.csv - Excel — D.Ranjan Sahoo

File | Home | Insert | Page Layout | Formulas | Data | Review | View | Help | Tell me what you want to do

Paste | Calibri | 11 | Wrap Text | General | Conditional Formatting | Format as Table | Cell Styles | Insert | Delete | Format | Sort & Filter | Find & Select

Clipboard | Font | Alignment | Number | Styles | Cells | Editing

POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited (.csv) format. To preserve these features, save it in an Excel file format. Don't show again | Save As...

A1 | class | Formula Bar

	A	B
1	class	message
2	ham	Go until jurong point, crazy.. Available only in bugis n great world la e buffet... Cine there got amore wat...
3	ham	Ok lar... Joking wif u oni...
4	spam	Free entry in 2 a wkly comp to win FA Cup final tkts 21st May 2005. Text FA to 87121 to receive entry question(std txt rate)T&C's apply 08452810075over18's
5	ham	U dun say so early hor... U c already then say...
6	ham	Nah I don't think he goes to usf, he lives around here though
7	spam	FreeMsg Hey there darling it's been 3 week's now and no word back! I'd like some fun you up for it still? Tb ok! XxX std chgs to send, å£1.50 to rcv
8	ham	Even my brother is not like to speak with me. They treat me like aids patent.
9	ham	As per your request 'Melle Melle (Oru Minnaminunginte Nurungu Vettam)' has been set as your callertune for all Callers. Press *9 to copy your friends Callertune
10	spam	WINNER!! As a valued network customer you have been selected to receivea å£900 prize reward! To claim call 09061701461. Claim code KL341. Valid 12 hours only.
11	spam	Had your mobile 11 months or more? U R entitled to Update to the latest colour mobiles with camera for Free! Call The Mobile Update Co FREE on 08002986030
12	ham	I'm gonna be home soon and i don't want to talk about this stuff anymore tonight, k? I've cried enough today.
13	spam	SIX chances to win CASH! From 100 to 20,000 pounds txt> CSH11 and send to 87575. Cost 150p/day, 6days, 16+ TsandCs apply Reply HL 4 info
14	spam	URGENT! You have won a 1 week FREE membership in our å£100,000 Prize Jackpot! Txt the word: CLAIM to No: 81010 T&C www.dbuk.net LCCLTD POBOX 4403LDNW1A7RW18
15	ham	I've been searching for the right words to thank you for this breather. I promise i wont take your help for granted and will fulfil my promise. You have been wonderful and a blessing at all times
16	ham	I HAVE A DATE ON SUNDAY WITH WILL!!
17	spam	XXXMobileMovieClub: To use your credit, click the WAP link in the next txt message or click here>> http://wap. xxxmobilemovieclub.com?n=QJKGIGHJJGCBL
18	ham	Oh k...i'm watching here:)
19	ham	Eh u remember how 2 spell his name... Yes i did. He v naughty make until i v wet.
20	ham	Fine if thatâÕs the way u feel. ThatâÕs the way its gota b

spam

6. TESTING

6.1 System Testing and Causes of Errors

System testing is an essential phase in the software development lifecycle that ensures the complete and integrated system meets the defined requirements. In the context of the Email Fraud Detection Using Machine Learning project, system testing evaluates the system's functionality, performance, usability, security, and reliability. It verifies that the machine learning model, web interface, database, and all system components work cohesively to deliver accurate predictions and a smooth user experience.

During testing, the system is subjected to various real-world scenarios, including the processing of diverse email datasets containing both legitimate and fraudulent messages. This helps ensure the system is resilient against edge cases and can handle exceptions effectively.

Causes of Errors in this project may stem from several sources:

- **Data-related issues:** Incomplete or incorrectly labeled training data can degrade model performance.
- **Model overfitting or underfitting:** Poor generalization of the model may result in incorrect classifications.
- **Code bugs:** Logic errors or syntax issues in the Python or Django codebase can hinder application performance.
- **Integration errors:** Miscommunication between frontend and backend components, or improper API usage, can lead to system crashes or incorrect outputs.
- **User interface flaws:** Unintuitive or broken UI elements can impact user interaction.
- **Deployment configuration:** Incorrect server or environment settings can cause failures in a production setting.

Understanding these potential sources of error is crucial for constructing robust test cases and improving system reliability.

6.2 Testing Objectives

The primary objectives of system testing for this project are:

1. **Functionality Verification:** Ensure that all functionalities, including input submission, model processing, and result output, work as expected.
2. **Performance Evaluation:** Assess the system's response time, especially under varying load conditions.
3. **Accuracy Assessment:** Validate that the machine learning model correctly classifies emails with a high precision and recall.
4. **Security Verification:** Ensure data entered by users is protected, especially against injection and cross-site scripting (XSS) attacks.
5. **User Interface Testing:** Confirm that the web interface is responsive, intuitive, and accessible across different devices and browsers.
6. **Error Handling Validation:** Check that the system gracefully handles unexpected inputs and errors without crashing.
7. **Integration Testing:** Validate that the frontend, backend, database, and machine learning model work together without conflict.

8. **Regression Testing:** Ensure that recent changes or enhancements do not break existing functionality.

These objectives help maintain a high-quality, user-centric product and ensure system stability post-deployment.

6.3 Kinds of Testing

Various types of testing were employed to ensure comprehensive evaluation of the system:

1. **Unit Testing:** Individual functions and methods were tested in isolation, especially those in the preprocessing and prediction pipeline. Python's unittest and pytest frameworks were used to test modules like TF-IDF vectorization and classifier predictions.
2. **Integration Testing:** This validated that individual modules such as the frontend, Django views, and ML model worked cohesively. Testing ensured that when an email was submitted, it flowed correctly through all components.
3. **System Testing:** Comprehensive testing of the entire application in a simulated production environment. This included end-to-end testing of user input, backend processing, database interaction, and result display.
4. **Functional Testing:** Confirmed that the application met the defined functional requirements, such as detecting email fraud, viewing predictions, and retraining models.
5. **Usability Testing:** Conducted with real users to gather feedback on interface design, ease of use, and response time.
6. **Performance Testing:** Assessed how the system handled high loads and stress scenarios using tools like Apache JMeter.
7. **Security Testing:** Ensured the system was protected from unauthorized access and common vulnerabilities using tools like OWASP ZAP.
8. **Regression Testing:** Performed after updates to confirm that new code changes did not disrupt existing functionalities.

Using multiple testing types helped ensure that the system was not only functional but also robust and secure.

6.4 Testing Techniques Used

Various techniques were applied to detect and fix issues effectively:

1. **Black-Box Testing:** Used extensively for functional and system testing. Testers validated inputs and outputs without considering internal code structure. For example, testing whether submitting a spam email correctly returns a "Fraudulent" classification.
2. **White-Box Testing:** Employed for unit testing internal code components, especially during development of the ML pipeline and Django views. Code coverage was measured using tools like coverage.py.
3. **Boundary Value Analysis:** Emails of varying lengths and token counts were tested to check model consistency and robustness.
4. **Equivalence Partitioning:** Different classes of inputs (e.g., spam, ham, gibberish, empty inputs) were tested to ensure that the system handles each group appropriately.
5. **Error Guessing:** Based on experience, test cases were designed to simulate common user mistakes like submitting empty forms, using HTML/JavaScript in email content, or injecting SQL code.
6. **Load Testing:** Simulated hundreds of concurrent email submissions to assess how the system

performs under load. This helped optimize server configurations and database queries.

7. **Cross-Browser Testing:** Ensured the web interface worked correctly in Chrome, Firefox, Edge, and mobile browsers.
8. **Automated Testing:** Selenium was used to automate GUI testing. It helped in quickly running test cases for user interactions.

By employing a blend of manual and automated techniques, the testing phase successfully ensured a high-quality, reliable product.

System testing plays a vital role in confirming that the Email Fraud Detection system is accurate, efficient, and user-friendly. By identifying the causes of errors early and employing a range of testing types and techniques, the system has been validated for robustness and reliability. The successful completion of the testing phase ensures that the final deployed application delivers a smooth experience to users while effectively identifying and mitigating fraudulent emails. Continued testing and monitoring post-deployment will further strengthen the system and ensure adaptability to new types of email fraud.

7. CONCLUSION

The Email fraud is one of the most pervasive and damaging forms of cybercrime in the modern digital age. As communication increasingly relies on email for both personal and business use, protecting users from fraudulent messages has become more critical than ever. This project, "Email Fraud Detection Using Machine Learning," addresses this pressing concern by leveraging the power of machine learning techniques to automatically identify and flag potentially malicious emails. Through careful planning, data collection, model training, system development, and testing, this project demonstrates how advanced computational methods can be used to enhance cybersecurity.

One of the key takeaways from the project is the importance of high-quality, labeled data in building an effective fraud detection system. The success of the machine learning model is closely tied to the diversity and comprehensiveness of the training dataset. By using techniques such as TF-IDF for feature extraction and algorithms like SVM or Logistic Regression for classification, the system achieves high accuracy in distinguishing between legitimate and fraudulent emails. Additionally, regular updates and retraining are essential to maintain effectiveness as fraudsters constantly evolve their tactics.

The development of the system using Python and Django provided a robust and scalable framework for integration between the machine learning model and the web-based user interface. This allowed users to input email content and instantly receive classification results. The backend ensured smooth handling of requests, secure data processing, and efficient model execution, while the frontend focused on providing a clean, intuitive experience for the user.

System testing played a critical role in ensuring the reliability and functionality of the final product. A variety of testing techniques—including unit testing, integration testing, performance testing, and security testing—were employed to validate all aspects of the system. These tests confirmed that the system can accurately detect fraudulent emails under different usage conditions and that it behaves correctly even when subjected to invalid or unexpected input.

Furthermore, the project's approach to system design, analysis, and implementation provides a replicable blueprint for future developments in similar domains. Conceptual models and architectural planning ensured that the project maintained a coherent and logical structure from inception through completion. The use of data flow diagrams and entity-relationship modeling helped visualize the internal interactions and data management strategies clearly.

In conclusion, this project successfully demonstrates that machine learning is a powerful tool in combating email fraud. By automating the process of detecting and flagging suspicious messages, it reduces human effort, enhances digital safety, and provides a scalable solution adaptable to

various contexts. While the system already exhibits high performance, future work could focus on incorporating deep learning models, analyzing image-based spam, and enhancing multilingual support. Additionally, further research into real-time detection, model interpretability, and user feedback mechanisms can significantly strengthen the reliability and adoption of such systems in real-world environments.

The fight against cyber threats is ongoing, and technological advancements like this project form an essential part of that defense. With continued innovation and rigorous testing, the Email Fraud Detection system can evolve into a comprehensive security layer, contributing meaningfully to global efforts in cybersecurity.

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8. FUTURE SCOPE

1. **Integration of Deep Learning Models:** Future improvements can include models like RNNs, LSTMs, or transformer-based models (e.g., BERT) to better understand the context and semantics of email content.
2. **Real-Time Detection Capabilities:** Implementing real-time scanning and alert systems will allow users to identify fraudulent emails as they arrive, reducing the risk of falling for scams.
3. **Multi-language and Regional Support:** Expanding the model's capabilities to support emails written in different languages and regional dialects would broaden its usability globally.
4. **Enterprise Integration:** Seamless integration with popular email platforms such as Gmail, Outlook, and enterprise mail servers would facilitate widespread adoption in corporate settings.
5. **User Feedback System:** Incorporating mechanisms for user feedback on predictions (e.g., marking false positives/negatives) will help retrain and improve the model over time.
6. **Browser Plugin or Extension:** Developing a plugin for browsers or email clients can enable end-users to get immediate fraud detection insights within their email interface.
7. **Multimodal Content Analysis:** Extending the system to analyze attachments, embedded images, and phishing links through image processing and link verification techniques will enhance detection capabilities.
8. **Automated Model Retraining:** Implementing pipelines for periodic retraining using new email datasets can help the model stay up to date with evolving fraud patterns.
9. **Enhanced Security Measures:** Adding more robust encryption and privacy protections to safeguard user data during prediction processing.
10. **Scalability Enhancements:** Optimizing backend architecture to support large-scale deployment and high user traffic for enterprise and public use cases.

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